



UNIVERSITY OF ESWATINI

FACULTY OF SCIENCE AND ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE

PROJECT PLAN

Project name: Hostel Access System

Team Members:

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1. Introduction

This project focuses on creating a system that controls access to student residential buildings using scannable cards. The idea came from noticing common problems in student housing, such as people entering buildings they are not assigned to and the difficulty of keeping track of who comes in and out.

In many cases, access is controlled manually or not strictly monitored, which can lead to security risks. This system aims to solve that by making the process automatic, faster, and more reliable.

Each student will have a card linked to their information. When they scan it at the entrance, the system will check if they are allowed to enter and either grant or deny access. It will also keep a record of entry attempts, which helps improve security and makes it easier to track activity when needed.

This system is expected to make student residences safer and more organized, while also making access easier and quicker for students. It provides a simple but effective solution to a real problem in residential management.

2. User Involvement

The main users of the system are:

- **Students** – scan cards to gain access
- **Administrators** – manage student data and monitor access

The client (school/residence management) will:

- Provide student details
- Define access rules
- Give feedback during development

3. Risks and Mitigation

Risk	Impact	Solution
System failure	No access control	Backup system
Lost/stolen cards	Unauthorized access	Deactivate cards
Incorrect data	Access errors	Regular updates
Poor connection	Delays in response	Offline backup or faster server

4. Standards, Guidelines and Procedures

- Use **GitHub** for version control and collaboration
- Follow basic **coding standards**
- Keep **clear documentation**
- Regular commits and updates
- Team communication for all major changes

5. Organization of the Project

The team is structured as follows:

- **Team Leader** – manages the project and approves changes
- **Software Designers** – plan system structure and flow
- **Developers** – build the system
- **Tester** – checks for errors and system performance

Each member contributes through GitHub by committing their work.

6. Project Phases

The system will be developed in the following phases:

1. **Planning** – Define the problem and objectives
2. **Analysis** – Understand system requirements
3. **Design** – Create system structure and workflow
4. **Implementation** – Develop the system
5. **Testing** – Identify and fix errors
6. **Final Review** – Prepare for submission

7. Requirements Analysis and Design

The system requires:

- Card scanning functionality
- Database to store student and access data
- Server to process requests

Design tools used:

- Flowcharts
- System diagrams

8. Implementation

The system will be implemented using:

- Programming language (e.g. Java or Python)
- Database (e.g. MySQL)
- GitHub for collaboration

9. Testing

Testing will include:

- Checking card scanning accuracy
- Verifying access decisions
- Testing system speed and reliability
- Fixing bugs found during testing

10. Resources

Resources required:

- Computers/laptops
- Internet connection
- Development software (IDE)
- GitHub account
- Scanner (or simulated input)

11. Quality Assurance

To maintain quality:

- Regular testing
- Code reviews
- Team discussions
- Fixing errors early

12. Change Management

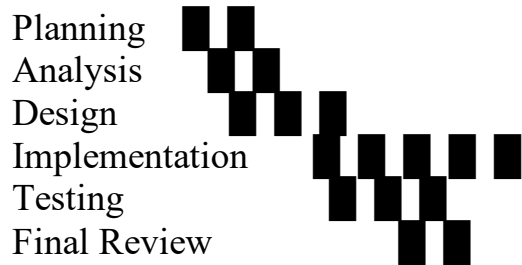
- All changes must be discussed with the team
- Changes must be committed on GitHub
- Updates must be documented

13. Task Planning Table

Task	Description	Duration (Days)	Dependency
Planning	Define problem & goals	2	-
Analysis	Gather requirements	2	Planning
Design	Create system structure	3	Analysis
Implementation	Develop system	5	Design
Testing	Test system	3	Implementation
Final Review	Prepare submission	2	Testing

14. Gantt Chart

Days → 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17



Conclusion

This project plan provides a clear structure for developing the system. It outlines tasks, responsibilities, and timelines, helping the team stay organized and complete the project successfully. Proper planning also reduces errors and ensures the system is secure and efficient.